

## Note 14.3 Partial Derivatives

### Partial Derivatives of a Function of Two Variables

Suppose  $(x_0, y_0)$  is a point in the domain of a function  $f(x, y)$ , then the vertical plane  $y = y_0$  will cut the surface  $z = f(x, y)$  as the curve  $z = f(x, y_0)$ . Here the  $y$ -value is constant at  $y_0$  so it is not a variable.

Then we can define the partial derivative of  $f$  with respect to  $x$  at the point  $(x_0, y_0)$  as the ordinary derivative of  $f(x, y_0)$  with respect to  $x$  at the point  $x = x_0$ .

#### Definition

The partial derivative of  $f(x, y)$  with respect to  $x$  at the point  $(x_0, y_0)$  is

$$\left. \frac{\partial f}{\partial x} \right|_{(x_0, y_0)} = \lim_{h \rightarrow 0} \frac{f(x_0 + h, y_0) - f(x_0, y_0)}{h}$$

if the limit exists. And  $h$  represents a real number. The symbol  $\partial$  is to distinguish the partial derivatives and the ordinary derivatives.

It can also be denoted by

$$\left. \frac{d}{dx} f(x, y_0) \right|_{x=x_0}, \frac{\partial f}{\partial x}(x_0, y_0), f_x(x_0, y_0), \left. \frac{\partial z}{\partial x} \right|_{(x_0, y_0)}, f_x, z_x, \frac{\partial f}{\partial x}, \frac{\partial z}{\partial x}$$

And the partial derivative of  $f(x, y)$  with respect to  $y$  is similar to that.

### Calculations

When finding  $\partial f / \partial x$ , we just need to see  $x$  as a variable and see  $y$  as a constant and vice versa.

### Functions of More Than Two Variables

The definitions and the methods are similar to the functions with two variables.

### Partial Derivatives and Continuity

Note that a function with two variables can have partial derivatives with respect to both  $x$  and  $y$  without the function being continuous there.

### Second-Order Partial Derivatives

Note the order when using different notations. Like

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x} \left( \frac{\partial}{\partial y} f \right)$$

but

$$f_{yx} = (f_y)_x$$

## The Mixed Derivative Theorem

**THEOREM 2—The Mixed Derivative Theorem** If  $f(x, y)$  and its partial derivatives  $f_x$ ,  $f_y$ ,  $f_{xy}$ , and  $f_{yx}$  are defined throughout an open region containing a point  $(a, b)$  and are all continuous at  $(a, b)$ , then

$$f_{xy}(a, b) = f_{yx}(a, b).$$

Note the theorem is **sufficient condition**.

## Exercises

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